Evaluating Password Security and User-Friendly Password Guidance

Student Research Proposal • Billal Hossain • Cybersecurity • Authentication • Usable Security

Proposed by	Billal Hossain
Project type	Student-led research / prototype study
Estimated duration	6–8 weeks

Project Overview

I want to study how password guidance can encourage stronger password choices without making authentication unnecessarily difficult for users. The project will compare different password-feedback strategies in a controlled prototype and examine the balance between security guidance and usability.

Student perspective
I want to keep the first version realistic for a student team: a focused research question, a reproducible experiment, and a prototype that can be discussed with a researcher and improved through collaboration.

Research Problem

Password policies can become difficult to follow when they contain many rules, yet very weak guidance may lead to predictable passwords. Instead of building another password generator, this project will study how security feedback is communicated and whether users can understand why a password is considered weak without exposing their actual passwords to a server.

Research Objectives

- Review common password-strength and password-policy principles.
- Design a local prototype that evaluates password characteristics without storing passwords.
- Compare different styles of strength feedback and guidance.
- Conduct a small usability study using synthetic or voluntarily entered test passwords.
- Document the trade-off between security recommendations and user experience.

Research Questions

RQ1. Which password feedback messages are easiest for users to understand?

RQ2. Can useful strength guidance be provided without storing or transmitting passwords?

RQ3. Which policy rules appear confusing or unnecessarily restrictive?

RQ4. How should a password-strength indicator communicate uncertainty and limitations?

Proposed Methodology

Step 1. Review usable-security literature and established password guidance.

Step 2. Define a local-only password evaluation prototype using non-reversible checks and safe test data.

Step 3. Create two or three feedback designs with different levels of explanation.

Step 4. Run controlled usability tasks with synthetic examples or voluntary participants.

Step 5. Measure comprehension, task completion, and user preference.

Step 6. Document security limitations and avoid treating a strength meter as proof of password safety.

Data / Materials / Experimental Resources

Local prototype, synthetic password examples, usability tasks, and a short anonymous questionnaire.

Evaluation Plan

Evaluate whether participants correctly understand the feedback and can select stronger example passwords. No real passwords will be collected or retained for analysis.

Expected Challenges

- Password strength cannot be perfectly measured from simple rules.
- Usability findings from a small student sample may not generalize widely.
- Feedback must avoid encouraging users to make predictable rule-based passwords.
- The prototype must never store or log actual passwords.

Reproducibility and Documentation

I will keep the main assumptions, dataset versions, preprocessing decisions, experiment settings, and evaluation results documented. The aim is that another student can understand how the result was produced and repeat the core experiment without relying on undocumented manual steps.

Planned Mitigation

I will begin with a small baseline and a feasibility check. If the original scope becomes too large for the available time or resources, I will narrow the research question rather than adding unnecessary complexity. Limitations and failed experiments will be documented as part of the research process.

Expected Outcomes

- A local password-feedback prototype.
- Comparison of alternative feedback designs.
- Small-scale usability findings.
- Security and usability recommendations for student authentication systems.

Potential Research Collaboration

I am looking for someone interested in cybersecurity, authentication, or human-computer interaction. A collaborator with usable-security experience could help design the study and interpret the findings.

Technical Stack

JavaScript, HTML, CSS, Node.js/Express for optional demo functionality, Git.

Scope, Ethics, and Responsible Research

The prototype will be designed so real passwords remain local to the user's browser and are not sent to a server. Participants will use synthetic or voluntarily created test examples, and no password database will be created.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Data/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, experiment design, implementation, analysis, or writing. I would like to divide responsibilities according to each person's strengths and keep the work transparent and documented.

Possible Next Step

If a researcher is interested, I would first like to discuss feasibility, refine the research question, and agree on a realistic division of work. The proposal can then be updated into a more specific study plan with milestones.

Note: This is a proposed student project. It does not claim completed research or guaranteed results, and the scope can be refined through collaboration.